

King Khalid University
College of Computer Science
Department of Computer Science



جامعة الملك خالد
كلية علوم الحاسب الآلي
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STUDENT LAB ACTIVITY REPORT

Course Name: Introduction to Machine Learning

Lab Teacher:

Course Code: 462 CCS-3

Student ID: 445801958

Section Number: 904/907

S. No

Student Name: Abdulmalik Hussein Al-Haif

LAB ACTIVITY - 5

(1.5 Marks)

Detecting Diabetics Using Logistic Regression

To improve the understanding of risk factors, we predict type 2 diabetes for Pima Indian women utilizing a logistic regression model and decision tree—a machine learning algorithm. Our analysis finds five main predictors of type 2 diabetes: glucose, pregnancy, body mass index (BMI), diabetes pedigree function, and age. We further explore a classification tree to complement and validate our analysis. The six-fold classification tree indicates glucose, BMI, and age are important factors, while the ten-node tree implies glucose, BMI, pregnancy, diabetes pedigree function, and age as the significant predictors.

Activity Requirements

Built the code by using the pima_indian_diabeties.csv file.

- 1) Load the data set and visualize the characteristics.
- 2) Normalize the data set.
- 3) Write the function to predict the output given the parameters computed by the GD.
- 4) Estimate logistic regression coefficients using stochastic gradient descent.
- 5) Plot the intercept vs. epochs. Plot the cost for the Sum of square errors vs. epochs.
- 6) Compute the prediction on the test data set by using the Logistic Regression
- 7) Compute the confusion matrix.
- 8) Visualize the model.

Dataset shape: (768, 9)

Normalized features using StandardScaler.

Training samples: 614

Test samples: 154

Logistic regression coefficients:

Pregnancies	0.363497
Glucose	1.188423
BloodPressure	-0.202243
SkinThickness	0.094903
Insulin	-0.130038
BMI	0.801103
DiabetesPedigreeFunction	0.318395
Age	0.178322

Accuracy on test data: 0.7078

Confusion matrix:

```
[[81 19]
 [26 28]]
```

Classification report:

	precision	recall	f1-score	support
0	0.76	0.81	0.78	100
1	0.60	0.52	0.55	54
accuracy			0.71	154
macro avg	0.68	0.66	0.67	154
Weighted avg	0.70	0.71	0.70	154

